$1 + p^a = q^b + r^c$

-7cm

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Background and Significance

Historical Context
The study of exponential Diophantine equations—where unknowns appear in exponents—has been a cornerstone of number theory. A celebrated example is Catalan's Conjecture ($x^a - y^b = 1$), proposed in 1844 and proved by Mihăilescu in 2002 using cyclotomic fields and linear forms in logarithms.

The Modulo Trick Method
In practice, many researchers first attempt to solve these equations using the **modulo trick**:

1. Choose a modulus m
2. Establish congruence: $1 + p^a \equiv q^b + r^c \pmod{m}$
3. This restricts possible values of exponents
4. Repeat with additional moduli until solutions are bounded

Brenner and Foster (1982) successfully applied this method to numerous equations of the form $1 + p^a = q^b + r^c$ with specific values of p, q, r.

The Gap
However, they remarked that certain classes:

Rationale for the Study

Why This Research Matters
The literature lacked:

- A **formal definition** of what it means for an equation to be "solvable with modulo trick"
- A **rigorous proof** of why certain equations resist modular treatment
- A **mathematical framework** to characterize when the method succeeds or fails

Research Questions

1. What does it mean for an equation to be "solvable by modulo trick"?
2. Why are some equations "non-solvable" by this method?
3. Can we prove the non-solvability of cases Brenner and Foster observed?

Our Approach
We bridge the gap between heuristic observations and formal impossibility proofs by:

- Introducing precise mathematical definitions
- Developing a formal framework for the modulo trick
- Proving non-solvability for specific equation families
- Validating prior research observations

Research Objectives

Main Goal: Turn the modulo trick from a useful idea into a precise method we can test, prove, and apply.

Objective 1: Define
Formalize "solvable with modulo trick" and "non-solvable with modulo trick" based on whether the method yields a bounded solution set.

Objective 2: Build
Develop a mathematical framework that models the modulo trick method using sets and congruences.

Objective 3: Prove
Establish rigorous proofs showing when the modulo trick method fails for specific equation families.

Methodology

Mathematical Framework Development
Step 1: Computability Proposition
We first establish when an equation is "computable" (all solutions accessible in finite time):

Proposition 1. *Equation $1 + p^a = q^b + r^c$ is computable if at least one condition holds:*
1. $\exists M$ such that $a \le M$ for any solution (a, b, c)
2. $\exists M_b, M_c$ such that $b \le M_b$ and $c \le M_c$ for any solution

Step 2: Define Solution Sets
Let $m_1, m_2, ..., m_k$ be moduli used. Define:

$$S_0^{p,q,r} = \mathbb{Z}^3_{\ge 0}$$

$$S_i^{p,q,r} = \{(a, b, c) \in S_{i-1}^{p,q,r} \mid 1 + p^a \equiv q^b + r^c \pmod{m_i}\}$$

Step 3: Key Lemma (LCM Property)

Lemma 1. *For any moduli $m_1, ..., m_k$:*

$$S_k^{p,q,r} = \bigcap_{i=1}^{k} A_{m_i} = A_{\mathit{lcm}(m_1,...,m_k)}$$

Significance: Any sequence of moduli reduces to a single modulus (LCM).

Step 4: Formal Definitions
Define "solvable" when finite moduli yield bounded solutions; "non-solvable" when unbounded solutions persist for any finite moduli choice.

Results - Theorem 1

Theorem 1 (Main Result)
For fixed coprime $p, q \in \mathbb{N} \setminus \{1\}$:

$$1 + (pq)^a = p^b + q^c$$

is **non-solvable with modulo trick**.

Proof Strategy:

1. Partition prime factors of any modulus m into sets for p, q, and neither
2. Factor $m = \chi \cdot \Gamma \cdot \mathrm{Z}$ where:
   - χ contains primes dividing p
   - Γ contains primes dividing q
   - Z contains primes dividing neither
3. Apply Euler's theorem with parameters $\theta_a, \theta_b, \theta_c$
4. Construct solutions with arbitrarily large exponents

Corollary: For distinct primes p, q, the equation is non-solvable with modulo trick.

Significance: This validates Brenner & Foster's comment (8.037) that $1 + (pq)^a = p^b + q^c$ "seems not amenable" to their method.

Results - Theorem 2

Theorem 2 (Generalization)
For $a, b \in \mathbb{Z}^+$:

$$a^x - a^y = b^z - b^w$$

is **non-solvable with modulo trick**.

Key Lemma (Period of Exponent Residues):

Lemma 2. *The sequence $(a^x \bmod n)_{x=1}^{\infty}$ is eventually periodic. That is, $\exists k, \epsilon \in \mathbb{Z}_{\ge 0}$ such that:*

$$a^x \equiv a^{x+\epsilon} \pmod{n} \quad \forall x \ge k$$

Proof: Uses pigeonhole principle on the finite set $\{0, 1, ..., n-1\}$.

Proof of Theorem 2:

1. For any modulus d, Lemma 8 gives periods ϵ_a, ϵ_b
2. For any $x \ge k_a, y \ge k_b$:

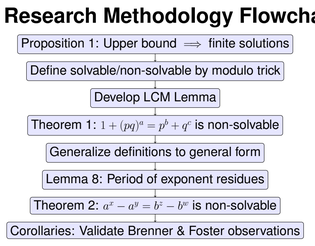
$$(x,\ x + N\epsilon_a,\ y,\ y + M\epsilon_b) \in A'_d$$

3. These can be made arbitrarily large

Corollary: For distinct primes p, q: $p^x - p^y = q^z - q^w$ is non-solvable (validates Brenner & Foster 8.033).

Methodology Flowchart

Research Methodology Flowchart:



Key Insight: The modulo trick succeeds exactly when the congruence conditions eventually force all solutions into a bounded set.

Conclusion and Summary of Findings

Summary of Contributions:

1. **First rigorous definitions** of "solvable/non-solvable with modulo trick"
2. **Mathematical framework** using solution sets $S_k^{p,q,r}$ and LCM lemma
3. **Theorem 1:** Proved $1 + (pq)^a = p^b + q^c$ is non-solvable for coprime p, q
4. **Theorem 2:** Proved $a^x - a^y = b^z - b^w$ is non-solvable for any a, b
5. **Corollaries** validating Brenner and Foster's heuristic observations

Practical Recommendations:
For equations of the forms:

- $1 + (pq)^a = p^b + q^c$
- $a^x - a^y = b^z - b^w$

Do NOT use the modulo trick. Instead, proceed directly to:

- Baker's theory of linear forms in logarithms
- Upper bounds from Tijdeman and Wang (1988)
- Advanced computational methods

Future Research Directions:

- Find non-trivial equations that ARE solvable with modulo trick
- Develop complete characterization of solvability conditions
- Apply framework to other forms of exponential Diophantine equations
- Investigate connections to Baker's theory

Acknowledgements

Acknowledgements:

Academic Support:

- Assoc. Prof. Dr. Kijti Rodtes (Research Advisor)
- Department of Mathematics, Faculty of Science
- Naresuan University

Institutional Support:

- Science Classrooms in University-Affiliated School Project (SCiUS)
- Naresuan University Secondary Demonstration School
- Research facilities and resources

Personal Support:

- Our families for unwavering support
- Friends at SCiUS for the wonderful environment
- All who contributed to this research

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Keywords

Keywords: Exponential Diophantine Equations · Modulo Trick · Modular Arithmetic · Non-solvability Proofs · Baker's Theory · Brenner and Foster